

```

# Import library
import pandas as pd
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report

# -----
# 1. Membuat DataFrame sesuai data pada gambar
# -----

data = {
    "Temperature": [14.0, 39.0, 30.0, 38.0, 27.0, 32.0, -2.0, 3.0, 3.0, 28.0],
    "Humidity": [73, 96, 64, 83, 74, 55, 97, 85, 83, 74],
    "Wind Speed": [9.5, 8.5, 7.0, 1.5, 17.0, 3.5, 8.0, 6.0, 6.0, 8.5],
    "Precipitation": [82.0, 71.0, 16.0, 82.0, 66.0, 26.0, 86.0, 96.0, 66.0, 82.0],
    "Cloud Cover": ["partly cloudy", "partly cloudy", "clear", "clear", "overcast", "overcast", "partly cloudy", "overcast", "clear", "partly cloudy"],
    "Atmospheric Pressure": [1010.82, 1011.43, 1018.72, 1026.25, 990.67, 1010.03, 990.87, 984.46, 999.44, 1012.13],
    "UV Index": [2, 7, 5, 7, 1, 2, 1, 1, 0, 8],
    "Season": ["Winter", "Spring", "Spring", "Spring", "Winter", "Summer", "Winter", "Winter", "Winter", "Winter"],
    "Visibility": [3.5, 10.0, 5.5, 1.0, 2.5, 5.0, 4.0, 3.5, 1.0, 7.5],
    "Location": ["inland", "inland", "mountain", "coastal", "mountain", "inland", "inland", "inland", "mountain", "coastal"],
    "Weather Type": ["Rainy", "Cloudy", "Sunny", "Sunny", "Rainy", "Cloudy", "Snowy", "Snowy", "Snowy", "Sunny"]
}

df = pd.DataFrame(data)
df

```

	Temperature	Humidity	Wind Speed	Precipitation	Cloud Cover	Atmospheric Pressure	UV Index
0	14.0	73	9.5	82.0	partly cloudy	1010.82	2
1	39.0	96	8.5	71.0	partly cloudy	1011.43	7
2	30.0	64	7.0	16.0	clear	1018.72	5
3	38.0	83	1.5	82.0	clear	1026.25	7
4	27.0	74	17.0	66.0	overcast	990.67	1
5	32.0	55	3.5	26.0	overcast	1010.03	2
6	-2.0	97	8.0	86.0	overcast	990.87	1
7	3.0	85	6.0	96.0	partly cloudy	984.46	1

Next steps:

[Generate code with df](#)[New interactive sheet](#)

Label Encoding untuk data kategori

```
# Label Encoding untuk kolom kategori
label_cols = ["Cloud Cover", "Season", "Location", "Weather Type"]
```

```
encoders = {}
for col in label_cols:
    enc = LabelEncoder()
    df[col] = enc.fit_transform(df[col])
    encoders[col] = enc
```

```
df
```

	Temperature	Humidity	Wind Speed	Precipitation	Cloud Cover	Atmospheric Pressure	UV Index	Season
0	14.0	73	9.5	82.0	2	1010.82	2	Summer
1	39.0	96	8.5	71.0	2	1011.43	7	Summer
2	30.0	64	7.0	16.0	0	1018.72	5	Summer
3	38.0	83	1.5	82.0	0	1026.25	7	Summer
4	27.0	74	17.0	66.0	1	990.67	1	Summer
5	32.0	55	3.5	26.0	1	1010.03	2	Summer
6	-2.0	97	8.0	86.0	1	990.87	1	Summer
7	3.0	85	6.0	96.0	2	984.46	1	Summer
8	3.0	83	6.0	66.0	1	999.44	0	Summer
9	22.0	71	2.5	107.0	0	1010.10	0	Summer

Next steps:

[Generate code with df](#)[New interactive sheet](#)

Split Data + Scaling

```
X = df.drop("Weather Type", axis=1)
y = df["Weather Type"]

# Scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Train Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.4, random_state=42, stratify=y
)
```

Tarining Model KNN

```
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
```

▼ KNeighborsClassifier ⓘ ?
KNeighborsClassifier(n_neighbors=3)

Evaluasi Model

```
from sklearn.metrics import accuracy_score, classification_report

# Prediksi
y_pred = knn.predict(X_test)

# Akurasi model
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# Report tanpa warning
report = classification_report(y_test, y_pred, zero_division=0)
print(report)
```

```
Accuracy: 0.75
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1
1	0.00	0.00	0.00	1
2	0.50	1.00	0.67	1
3	1.00	1.00	1.00	1
accuracy			0.75	4
macro avg	0.62	0.75	0.67	4
weighted avg	0.62	0.75	0.67	4

Penjelasan

Model berhasil memprediksi 3 dari 4 data uji dengan benar sehingga menghasilkan akurasi 75%. Performa model bervariasi antar kelas; beberapa kelas diprediksi dengan sangat baik, sementara kelas lain tidak berhasil diprediksi sama sekali akibat ukuran dataset yang kecil dan distribusi kelas yang tidak merata.

Kesimpulan

Model KNN berhasil mencapai akurasi sebesar 75% dalam memprediksi jenis cuaca. Hasil precision, recall, dan f1-score menunjukkan bahwa model bekerja sangat baik pada beberapa kelas namun lemah pada kelas tertentu karena keterbatasan jumlah data. Untuk meningkatkan performa, disarankan menggunakan dataset yang lebih besar dan mencoba optimasi nilai K.